

CONTACT

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EDUCATION

M.Sc. AI and Robotics
Hochschule Hof, Bavaria
Mar 2025 – ongoing

Applied Deep Learning, Generative AI, Advanced AI Applications, Industry 4.0 / Data Management, Predictive Maintenance and Condition Monitoring.

B.E. Information Technology
SCET (GTU), Surat, India
2019 – 2023 · GPA 1.8

CORE SKILLS

Python · PyTorch · Hugging Face · Transformers · LLM Fine-Tuning · SFT · LoRA / QLoRA · DPO · RLHF · RAG · LangChain · LlamaIndex · FAISS · BM25 · Reranking · NLP · TRL · MLflow · FastAPI · Docker · Evaluation

LANGUAGES

German — CEFR B2
English — Fluent
Hindi — Native
Gujarati — Native

Prit Dhanani

AI Engineer — LLM Fine-Tuning, Retrieval-Augmented Generation & Evaluation

SUMMARY

AI Engineer specialising in adapting large language models to narrow, high-stakes domains — supervised fine-tuning, DPO preference alignment and retrieval-augmented generation, built and measured end to end. Working Student at INFINITE Mind GmbH in Munich, shipping on-premise GenAI systems under strict privacy constraints. Seeking a 6-month Master's thesis (Abschlussarbeit) from October 2026 to build, compare and systematically evaluate RAG and fine-tuning approaches on real engineering knowledge sources.

EXPERIENCE

Generative AI Engineer (Working Student)
Sep 2025 – Present
INFINITE Mind GmbH, Munich

- Designed and deployed enterprise-grade GenAI systems (RAG pipelines, chatbots, agentic workflows) using LangChain, LlamaIndex and FastAPI, improving information retrieval efficiency by 35%.
- Built on-premise LLM adaptation and RAG pipelines that run without external APIs, meeting enterprise privacy and data-residency constraints.
- Built an LLM-powered agentic system for automated content and business-proposal generation using LangChain and Python, reducing manual production time by 95% (hours to under 2 minutes).
- Orchestrated the end-to-end delivery lifecycle from requirement gathering to deployment, delivering 3 production-ready systems to enterprise clients.

AI Engineer — Woxxin Solution Pvt. Ltd, India
Oct 2023 – Jul 2024

- Fine-tuned open-source LLMs using PyTorch and LoRA on domain-specific datasets, increasing chatbot response accuracy by 22%.
- Managed the complete lifecycle of client-facing AI products as sole owner, using Azure AI and Azure OpenAI to deliver custom enterprise chatbot solutions.
- Developed and productionised predictive ML models with TensorFlow and Scikit-learn using Docker and CI/CD pipelines, shortening deployment cycles by 30%.
- Designed automated data ingestion and preprocessing pipelines, using MLflow for real-time performance tracking and drift detection.

SELECTED PROJECTS — LLM ADAPTATION, RAG & EVALUATION

Local LLM Training & Optimization Suite — On-Premise SFT and DPO Alignment

- Engineered a privacy-preserving local training platform enabling SFT and DPO alignment of 7B/8B models (e.g. Llama 3) on a single 16 GB VRAM GPU, reducing GPU memory footprint by 60% through NF4/QLoRA quantisation, activation checkpointing and paged optimizers.
- Implemented subprocess-level process isolation in PyTorch and FastAPI so 100% of CUDA VRAM is reclaimed by the OS on job completion, eliminating CUDA context locks; increased training throughput by 2.4x.

CERTIFICATIONS

AWS Certified AI Practitioner
Building Cloud-Native and Multicloud Applications (IBM)
Machine Learning with Python
Python for Data Science

AWARDS

ML Hackathon 2.0 — 2nd Place
2023

MYSY Scholarship
Outstanding academic progress, 2019–2023

TOOLS & PLATFORMS

TensorFlow · Keras · Scikit-learn · XGBoost · Ollama · CrewAI · W&B · DVC · Kubernetes · GitHub Actions · GitLab CI · PostgreSQL · SQL · Pandas · NumPy · AWS · SageMaker · Bedrock · Azure ML · Azure OpenAI · Streamlit · Django · React · CUDA

AVAILABILITY

Master's thesis (Abschlussarbeit)
6 months, full-time
Earliest start: October 2026
Location: Munich

CURRENT FOCUS

On-premise alignment pipelines under privacy constraints
LLM evaluation frameworks and reference benchmarks
Multi-agent orchestration
Reasoning in resource-constrained environments

- Enforced a strict prompt / chosen / rejected preference-data schema with de-duplication and minimum-length filtering, and streamed loss, learning-rate and GPU metrics to a React dashboard with sub-100 ms updates.

Tech: PyTorch, TRL (SFT/DPO), QLoRA/NF4, Hugging Face Transformers, FastAPI, React, SQLite, CUDA.

Production RAG Platform — Hybrid Retrieval, Reranking & Real-Time Evaluation

- Engineered a RAG pipeline combining BM25 sparse search, FAISS dense vector retrieval, query expansion and a cross-encoder reranking layer, improving retrieval accuracy by 35% and reducing hallucinations.
- Applied prompt compression to optimise context, decreasing token payload by 40% and accelerating end-to-end inference speed.
- Designed an ingestion pipeline that parses, cleans and semantically chunks heterogeneous knowledge sources (PDF, Markdown, HTML, crawled documentation) into a single unified index.
- Integrated an MLflow observability layer tracking retrieval similarity scores, query latency and answer faithfulness against reference queries in real time.

Tech: LlamaIndex, LangChain, FAISS, BM25, sentence-transformers, FastAPI, Docker, MLflow, PostgreSQL.

Deck Engine — Schema-Constrained, Verifiable LLM Generation (v1.0.0)

- Designed a generation architecture that separates deterministic logic from model output: layout selection, chart-type resolution and template cloning are resolved in code, while the LLM supplies only text and data.
- Enforced per-field constraints at three layers — injected into the prompt, enforced as a JSON schema, and re-validated after generation — with an "omit, don't invent" rule so the model leaves fields empty rather than fabricating content.
- Implemented a bounded retry loop that returns exact validation errors to the model and escalates an item for human review after repeated failure; abstracted three inference backends (Anthropic, Gemini, local Ollama) behind a single interface.

Tech: Python 3.13, Pydantic, JSON Schema, python-pptx, Anthropic and Gemini APIs, Ollama.

RESEARCH & ACADEMIC WORK

- Conducted a systematic review of Knowledge Graph Embeddings (KGE), evaluating advanced neural and hyperbolic models (KG-BERT, HypHKGE) to identify key performance benchmarks for semantic search.
- Implemented the Auto-Encoding Variational Bayes (AEVB) algorithm directly from the original research paper, creating a PyTorch prototype for variational autoencoders (VAEs) with 95% reconstruction accuracy.
- Analysed edge AI architectures at the Siemens Amberg Plant, proposing LLM-integrated factory workflows to optimise the plant's 75% manufacturing automation.
- SmartDiet AI (team project): identified and documented near-deterministic label leakage in a public health dataset that produced perfect classifier scores, and reported realistic clinical baselines (70–85%) instead of the inflated metric.